

PULLARI SHIVA KUMAR YADAV

AI / Machine Learning Engineer

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PROFESSIONAL SUMMARY

Motivated AI and Machine Learning enthusiast currently pursuing a B.Tech in Electronics & Communication Engineering (Gen AI Specialization) at KLH University, Hyderabad, with a current CGPA of 9.57. Hands-on experience in Python-based Computer Vision during an industry internship. Proficient in Deep Learning, NLP, Computer Vision, and Generative AI including Agentic AI, Retrieval-Augmented Generation (RAG), and transformer-based models. Independently built multiple end-to-end AI projects using PyTorch, TensorFlow, Scikit-learn, FastAPI, and YOLOv8. Secured ₹1,00,000 in grant funding from IIT Hyderabad's TiHAN to develop an agentic, AI-driven Intelligent Transportation Management System (ITMS). Won 5 college-level hackathons. Passionate about applying AI to solve real-world challenges in cybersecurity, intelligent automation, and data analytics.

TECHNICAL SKILLS

- **Programming:** Python, SQL / MySQL, C, Java (Learning)
- **Data Science & ML:** Data Science, Scikit-learn, TensorFlow / Keras, PyTorch, CNNs & Deep Learning, NLP & Transformers, YOLOv8, Computer Vision
- **Gen AI & LLMs:** Agentic AI, RAG Systems, LangChain, Prompt Engineering, Transformer Models, Generative AI
- **Tools, Data & Deployment:** Pandas / NumPy, Matplotlib / Seaborn, Git & GitHub, Jupyter / Colab, FastAPI, Docker, Microsoft Excel

WORK EXPERIENCE

Python Developer Intern — Computer Vision

Feb 2025 – Apr 2025

Robosevak, Hyderabad, India

- Developed and implemented computer vision pipelines in Python for robotic perception and navigation tasks.
- Worked with image preprocessing, object detection, and real-time inference modules integrated with robotic systems.
- Collaborated with the engineering team to optimize model performance and reduce inference latency.
- Gained hands-on experience applying YOLOv8 and OpenCV in an industry robotics environment.

PROJECTS

Intelligent Transportation Management System (ITMS)

2025 – 2026 (Ongoing)

★ FUNDED PROJECT — ₹1,00,000 GRANT, IIT HYDERABAD (TiHAN)

- Secured ₹1,00,000 in funding from IIT Hyderabad's TiHAN (Technology Innovation Hub on Autonomous Navigation) to design and build an AI-driven Intelligent Transportation Management System.
- Building agentic AI workflows so autonomous agents can handle traffic monitoring, incident detection, and route/signal optimization with minimal human intervention.
- Architecting a multi-agent system that integrates perception, decision-making, and control components for real-world transportation scenarios.

Smart Recruitment Assistant using BERT

2025 – 2026

- Built an AI-powered recruitment assistant using pretrained BERT embeddings for resume screening and candidate-job matching.
- Implemented semantic similarity and NLP pipelines to rank candidates based on job descriptions.
- Applied transformer-based tokenization and embeddings for contextual understanding.
- Explored RAG concepts for intelligent information retrieval.

Chatbot with MySQL — Embeddings & Reranking (RAG)

2025 – 2026

- Built a retrieval-augmented chatbot that answers natural-language questions over data stored in MySQL.
- Implemented embedding-based semantic search combined with a reranking layer to surface the most relevant records before generating a response.

- Designed the pipeline to convert user queries into embeddings, retrieve candidate matches from MySQL, rerank them for relevance, and feed the top results to an LLM for final response generation.

Image Enhancement & Object Detection using YOLOv8

2025 – 2026

- Developed a computer vision pipeline with denoising and contrast enhancement before YOLOv8-based inference.
- Evaluated model performance using mAP, precision, and recall metrics.

Shopping Agent — VLM + LLM Powered Assistant

2025 – 2026

- Developed an AI shopping agent, similar in spirit to the AI shopping assistant inside the Amazon app, that combines a Vision-Language Model (VLM) with an LLM to understand product images and natural-language requests.
- The agent interprets user intent from text and images, reasons over available products, and autonomously performs actions such as searching, comparing, and updating the cart.
- Implemented the agent to read from and write to a SQLite database directly, executing real changes (e.g., cart and order updates) based on its own decisions.

Night Eye — AI-Based Drone Vision Enhancement

Ongoing

- Building an AI pipeline for enhancing low-light drone imagery using deep learning techniques.
- Integrating semantic segmentation and YOLO-based detection for terrain and obstacle identification.

Underground Worker Safety Tracking System (No GPS)

2025 – 2026

- Designed an AI + IoT-based safety tracking system for underground environments using sensor-based positioning.
- Integrated real-time monitoring workflows for safety alerts and worker status tracking.

EDUCATION

B.Tech — Electronics & Communication Engineering (Gen AI Specialization)

2024 – 2028

KLH University, Hyderabad — CGPA: 9.57 — Currently in 3rd Year

Relevant Coursework: Machine Learning, Deep Learning, Python Programming, Data Structures & Algorithms, DBMS, Data Analytics, Computer Vision, NLP Fundamentals

CERTIFICATIONS

- Oracle Cloud Infrastructure Data Science Professional — Oracle, 2025
- Python Intermediate Certification — HackerRank, 2025
- MySQL Intermediate Certification — HackerRank, 2025
- AWS Certified Cloud Practitioner (CLF-C02) — Amazon Web Services (AWS), 2026
- MongoDB Certified Developer Associate — MongoDB, 2026

ACHIEVEMENTS

- Won 5 College-Level Hackathons focused on AI, Machine Learning, and Software Development.
- Secured ₹1,00,000 in grant funding from IIT Hyderabad's TiHAN for an agentic, AI-driven Intelligent Transportation Management System (ITMS).
- Built multiple end-to-end AI and data analytics projects independently, spanning NLP, Computer Vision, and Generative AI.
- Active learner in Agentic AI, Generative AI, LLMs, LangChain, and applied ML workflows.